

Pavement Crack Detection Using Otsu Thresholding for Image Segmentation

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Abstract—Pavement cracks are the first signs of structural damage in the asphalt pavement surfaces. The oldest method for detection and estimation of the pavement cracks is human visual inspection, also known as manual visual inspection. However, using human inspectors is very time consuming, very expensive and poses a risk to human safety. Another negative side is the fact that the task generally requires road to be closed. Hence, automatic prevention and reparation of cracks on the asphalt surface pavements is an important task, especially because the advanced stages of road deformation lead to formation of potholes. This has negative impact on the total reparation cost. In this paper, we proposed a new unsupervised method for the detection of cracks with gray color based histogram and Otsu's thresholding method on 2D pavement image. At first, the method divides the input image into a four independent equally sized sub-images. Then, the search for cracks is based on the ratio between Otsu's threshold and the maximum histogram value for every sub-image. Finally, all sub-images are assembled into the resulting image. The method was tested on the dataset which contains different pavement images with very versatile types of cracks. The results showed that the proposed method achieves satisfactory performance, especially in the cases of low signal-to-noise ratio, and is very fast.

Keywords: Cracks detection, Unsupervised method, Image processing, Image segmentation, Computer Vision

I. INTRODUCTION

Image-based pavement health monitoring techniques have a history of more than 30 years [1]. Techniques range from early windshield surveying to manual and/or computer aided analysis of pavement surfaces digital images. Currently, it is common practice to capture and analyze 2D digital images with the high-speed cameras mounted on a specialized or passenger vehicles. The analysis is usually performed off-line on an as-needed basis, or online to obtain current information. Today, many pavement maintenance agencies employ the semi-automated and the fully-automated image-based methods for collection of the pavement state data. For example, the national roads in France are periodically inspected in a three years cycle [2], while in Germany the relevant surface characteristics are inspected in a four year cycle [3].

The analysis of pavement images should reveal various existing distresses on asphalt pavement surfaces. They are usually classified as high- or low-level severity distresses. Former are commonly known as potholes [4]–[6] and cracks, while latter are patches [7], surface deformations and defects. Cracks on asphalt pavement surfaces are the most common distress type. They represent important indication of possible structural damage that may lead to dangerous situations.

Many algorithms with different approaches exist in literature [1], [8]–[11], however the advancements in development of accurate, automated, low-cost, effective and reliable image-based pavement cracks detection are still needed.

2D image analysis of any type of distress is a challenging task due to limiting range of information that can be extracted from 2D space. Cracks on images appear as thin, irregular, darker intensity lines, surrounded by strong textured noise. Technologies such as 3D imaging and texture analysis can provide additional information, but they are considered expensive in two ways: use of expensive equipment and very high demand for computational processing. Many supervised algorithms exist in literature [12]–[17], however very few fully unsupervised [18], [19] methods exist.

In this paper, we address the problem of efficient automated pavement cracks detection. We use unsupervised vision-based approach to analyze photometric data from 2D asphalt surface pavements images and extract important information to perform efficient image segmentation. The method consists of two major phases. In the first phase, the original image is sliced into a four sub-images, then arithmetic mean and standard deviation of each sub-image is calculated. In the second phase, we calculate histogram and Otsu's thresholding for each sub-image. Then, we define the smallest ratio between minimal and maximal absolute difference between these two values for all four sub-images. This information is then used to eliminate the background pixels and effectively select pixels that represent cracks. We have performed the comprehensive tests on the real pavement images with cracks at different severity levels. Tests show promising results with very low signal-to-noise ratio and very fast execution time. The method is applied without pre- or post-filtering and without any knowledge about shapes of cracks. The method is suitable for parallel execution since it can take advantage of spatial distribution of crack pixels and it can analyze four sub-images in parallel.

This paper is organized as follows: the related work is briefly reviewed in Section II. In Section III, a new method for detection of high severity distresses on the asphalt pavements has been proposed. The implementation details, data set and experimental results have been presented in Section IV. The paper is concluded with some remarks in Section V.

II. RELATED WORK

There is a substantial number of methods related to cracks detection in literature. In general, they can be classified into four approaches [8]: 1) photometry-based, 2) geometry constraints-based, 3) photometry- and geometry-based, and 4) minimal path-based approaches. We briefly review all of them to provide some background information.

Photometry-based approaches are based on the photometrical information extracted from the 2D color and spatial information. They usually involve determining some threshold value with simple thresholding operation. The threshold value can be determined based on the global [20] or the local image information with adaptive thresholding [21]. Threshold value can be determined based on Otsu's thresholding method, fuzzy logic or genetic algorithms. Majority of photometry-based approaches are unsupervised, however they can contain some supervised learning step, such as is described in [12]–[14], [22], [23].

Geometry constraints-based approaches include topology information at the local scale. There are various methods based on multi-scale analysis for the fine structure extraction. Two examples are based on the watershed method [24] and the wavelet decomposition [15].

Photometry- and geometry-based approaches combine photometric and topology information to represent cracks as contiguous pixels [2], [25]. This approach is considered as computationally intensive, since it relies on the analysis of statistical information present in the image. It also relies on the statistical parameter tuning, which usually requires some supervised knowledge.

Minimal path-based approaches employ existing Dijkstra's and the fast marching algorithm to exploit the high-level geometrical information, while maintaining processing time and memory usage. Even though these approaches are computationally efficient, it is difficult to implement them in an unsupervised manner, because some information (such as location of two endpoints for each path) are required in advance. Nevertheless, several methods [16], [26]–[28] suggest ways to overcome this problem. They all follow the common overall structure: 1) select the darkest pixels belonging to the crack structure, 2) select pairs of the darkest pixels as endpoints and build the minimal path between them. Some methods require manual input, while others are more automatic.

The comprehensive review and analysis of recent crack detection methods is published in [29].

III. METHODOLOGY

The overall design of our method is shown in Fig. 1. There are two major phases in the method: a) Image Slicing and b) Cracks Detection. We describe each phase in dedicated subsection.

A. Image Slicing

At first, the original image I is divided into four equal sub-images $S^{(i)}$ as described in equation (1), where $i \in \{1, 2, 3, 4\}$. Four new sub-images are formed assuming conditions in equation (2), where $p(x, y)_{[R,G,B]}^{(i)}$ represents pixel values of (x, y) location for RGB color components of sub-image $S^{(i)}$

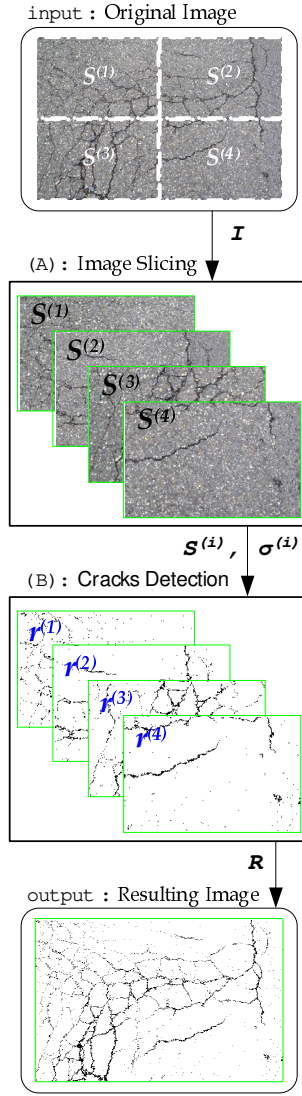


Figure 1: Block diagram of proposed method for cracks detection.

and $P(X, Y)_{[R,G,B]}$ represents pixel values of (X, Y) location for RGB color components of the original image I .

$$S^{(i)} = p(x, y)_{[R,G,B]}^{(i)} \quad (1)$$

$$\begin{aligned} p(x, y)_{[R,G,B]}^{(i)} &\subset P(X, Y)_{[R,G,B]} \\ (x, y) &\subset (X, Y) \end{aligned} \quad (2)$$

The arithmetic mean μ and the standard deviation σ are calculated by equation 3 and 4, respectively. This information is later used for image segmentation.

$$\mu^{(i)} = \frac{1}{N \times M \times 3} \sum_x \sum_y \sum_{[R,G,B]} p(x, y)_{[R,G,B]}^{(i)} \quad (3)$$

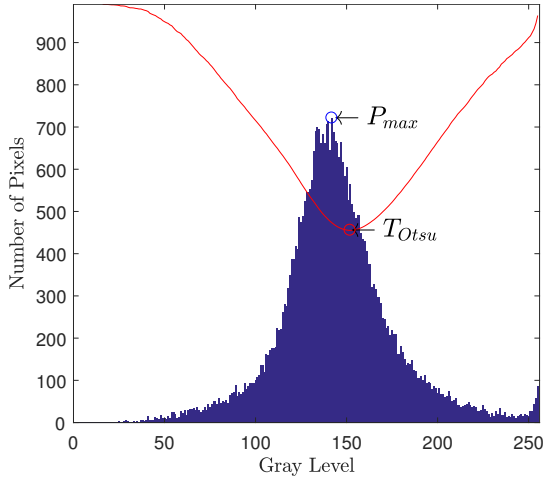


Figure 2: Histogram and Otsu's threshold for the sub-image $I^{(i)}$.

$$\sigma^{(i)} = \sqrt{\frac{1}{3NM-1} \sum_x \sum_y \sum_{[R,G,B]} [p(x,y)_{[R,G,B]}^{(i)} - \mu^{(i)}]^2} \quad (4)$$

Parameters N , M and $[R, G, B]$ denote height, width and color information of the $S^{(i)}$ sub-image, respectively.

The resulting $S^{(i)}$ and $\sigma^{(i)}$ are passed to the next phase.

B. Cracks Detection

In cracks detection phase, each sub-image $S^{(i)}$ is converted into gray scale image $I^{(i)}$. Then, histogram and Otsu's threshold is calculated for each sub-image $I^{(i)}$, as shown in Fig. 2. The value T_{Otsu} is Otsu's threshold, while P_{max} is the value of the most frequent pixel in the sub-image.

Detection of cracks on an image requires efficient elimination of background pixels. Hence, we define a set of values in equation (5), (6) and (7). Equation (5) describes ratio between minimal and maximal absolute difference between $T_{Otsu}^{(i)}$ and $P_{max}^{(i)}$ of all four sub-images. The values $T_{Otsu}^{(i)}$ and $T_{Otsu}^{(j)}$ are Otsu's thresholds, while $P_{max}^{(i)}$ and $P_{max}^{(j)}$ are gray levels of the most frequent pixels in the sub-images i and j , respectively. The resulting ϵ is the smallest value between the four sub-images.

$$\epsilon = \frac{\min |T_{Otsu}^{(i)} - P_{max}^{(i)}|}{\max |T_{Otsu}^{(j)} - P_{max}^{(j)}|}, \quad \forall i, j = \{1, 2, 3, 4\} \quad (5)$$

Next, we define modified value $\hat{\epsilon}$ to eliminate ϵ values greater than $1/2$ in equation (6). The result is less noise around cracks, since more pixels surrounding cracks are marked as background pixels. Cracks appear thinner and more accurate. This has shown promising results in cases when asphalt pavement is partially covered with water (i.e. after the rainy weather conditions).

$$\hat{\epsilon} = \begin{cases} 1/2, & \epsilon > 1/2 \\ \epsilon, & \text{otherwise} \end{cases} \quad (6)$$

In equation (7), we define new constant value δ for image segmentation. The equation was defined based on our test dataset (Section IV.B), however it showed promising results when applied to similar images.

$$\delta = \hat{\epsilon} + \begin{cases} 1.20, & |T_{Otsu}^{(i)} - P_{max}^{(i)}| > 30 \\ 1.17, & 30 \geq |T_{Otsu}^{(i)} - P_{max}^{(i)}| > 1 \\ 1.15, & \text{otherwise} \end{cases} \quad (7)$$

Image segmentation is described for two distinct cases: 1) for $T_{Otsu} > P_{max}$ defined in equations (8) and (9), and 2) for $P_{max} > T_{Otsu}$ defined in equations (10) and (11). Image segmentation is performed by a number of conditions. The majority of conditions require previously described values, such as δ , σ , T_{Otsu} and P_{max} .

The selection of background and cracks pixels is carried out in two passes. In the first pass, the values $g_{(x,y)}^{(i)}$ and $h_{(x,y)}^{(i)}$ describe initial selection of background and cracks pixels. This pass selects majority of pixels. In the second pass, the values $\hat{g}_{(x,y)}^{(i)}$ and $\hat{h}_{(x,y)}^{(i)}$ select remaining pixels for both cases.

$$g_{(x,y)}^{(i)} = \begin{cases} 0, & P_{max}^{(i)} < 10 \wedge I_{(x,y)}^{(i)} < 2 \cdot P_{max}^{(i)} \cdot \delta / \sigma^{(i)} \\ & \cdot (T_{Otsu}^{(i)} - P_{max}^{(i)}) \\ 0, & T_{Otsu}^{(i)} - P_{max}^{(i)} \geq 10 \wedge \\ & I_{(x,y)}^{(i)} < (T_{Otsu}^{(i)} - P_{max}^{(i)}) \cdot \delta \\ 0, & T_{Otsu}^{(i)} - P_{max}^{(i)} < 10 \wedge \\ & \sigma^{(i)} / (T_{Otsu}^{(i)} - P_{max}^{(i)}) > 6 \wedge \\ & \sigma^{(i)} > 30 \wedge I_{(x,y)}^{(i)} < \delta \cdot \sigma^{(i)} \\ 0, & T_{Otsu}^{(i)} - P_{max}^{(i)} < 10 \wedge \\ & I_{(x,y)}^{(i)} < (T_{Otsu}^{(i)} - P_{max}^{(i)}) \cdot \delta \wedge \\ & (\sigma^{(i)} / (T_{Otsu}^{(i)} - P_{max}^{(i)}) \leq 6 \vee \sigma^{(i)} \leq 30) \\ 0, & I_{(x,y)}^{(i)} \geq (T_{Otsu}^{(i)} - P_{max}^{(i)}) \cdot \delta \wedge \\ & I_{(x,y)}^{(i)} < T_{Otsu}^{(i)} / \delta \end{cases} \quad (8)$$

$$\hat{g}_{(x,y)}^{(i)} = \begin{cases} 1, & I_{(x,y)}^{(i)} > T_{Otsu}^{(i)} \\ g_{(x,y)}^{(i)}, & P_{max}^{(i)} < 128 \\ 0, & I_{(x,y)}^{(i)} < T_{Otsu}^{(i)} / \delta \wedge P_{max}^{(i)} \geq 128 \\ 1, & \text{otherwise} \end{cases} \quad (9)$$

$$h_{(x,y)}^{(i)} = \begin{cases} 0, & T_{Otsu}^{(i)} < 10 \wedge (I_{(x,y)}^{(i)} < 2 \cdot T_{Otsu}^{(i)} \\ & \cdot \delta \cdot 1/\sigma^{(i)} \cdot |T_{Otsu}^{(i)} - P_{max}^{(i)}| \vee \\ & I_{(x,y)}^{(i)} < (P_{max}^{(i)} - T_{Otsu}^{(i)}) \cdot \delta) \\ 0, & P_{max}^{(i)} - T_{Otsu}^{(i)} \geq 10 \wedge \\ & I_{(x,y)}^{(i)} < (P_{max}^{(i)} - T_{Otsu}^{(i)}) \cdot \delta \\ 0, & I_{(x,y)}^{(i)} \geq (P_{max}^{(i)} - T_{Otsu}^{(i)}) \cdot \delta \wedge \\ & I_{(x,y)}^{(i)} < P_{max}^{(i)} / \delta \end{cases} \quad (10)$$

$$\hat{h}_{(x,y)}^{(i)} = \begin{cases} 1, & I_{(x,y)}^{(i)} > P_{max}^{(i)} \\ h_{(x,y)}^{(i)}, & T_{Otsu}^{(i)} < 128 \\ 0, & I_{(x,y)}^{(i)} < T_{Otsu}^{(i)} / \delta \wedge P_{max}^{(i)} \geq 128 \\ 1, & \text{otherwise} \end{cases} \quad (11)$$

The pseudocode of our algorithm is described in Algorithm 1. The resulting image $r_{(x,y)}^{(i)}$ (lines 14 and 16) depends on the values T_{Otsu} and P_{max} . For the case when $T_{Otsu} > P_{max}$, the resulting image is $\hat{g}_{(x,y)}^{(i)}$, and for the case when $T_{Otsu} < P_{max}$, the resulting image is $\hat{h}_{(x,y)}^{(i)}$.

In the final step, four resulting sub-images $r^{(i)}$ are assembled into one resulting image R .

Algorithm 1 Cracks detection

Require: I - input RGB image

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1: for  $i \leq 4; i \leftarrow i + 1$  do
2:    $S^{(i)} \leftarrow \text{Equation (1)}$   $\triangleright$  Sub-image  $S^{(i)}$ 
3:    $\sigma^{(i)} \leftarrow \text{std}(S^{(i)})$   $\triangleright$  Standard deviation of the RGB
    sub-image  $S^{(i)}$  defined by equation (4).
4:    $I^{(i)} \leftarrow \text{rgb2gray}(S^{(i)})$   $\triangleright$  Conversion from RGB to
    grayscale.
5:    $[cntOfPix, p] \leftarrow \text{histogram}(I^{(i)})$ 
6:    $P_{max}^{(i)} \leftarrow \arg \max_p (cntOfPix)$   $\triangleright$  Parameter  $cntOfPix$  is
    number of pixels per gray level  $p$  of sub-image  $I^{(i)}$ .
7:    $T_{Otsu}^{(i)} \leftarrow \text{Otsu}(I^{(i)})$   $\triangleright$  Otsu threshold for the sub-image
     $I^{(i)}$ .
8: end for
9:
10:  $\delta \leftarrow \text{Eq. 7}$   $\triangleright$  Calculation of  $\delta$  using equation (7).
11:
12: for  $i \leq 4; i \leftarrow i + 1$  do  $\triangleright i$  denotes current sub-image  $I^{(i)}$ .
13:   if  $T_{Otsu}^{(i)} > P_{max}^{(i)}$  then
14:      $r_{(x,y)}^{(i)} \leftarrow \hat{g}_{(x,y)}^{(i)}$   $\triangleright$  The case when  $T_{Otsu}^{(i)} > P_{max}^{(i)}$ 
    defined in equation (8) and (9).
15:   else
16:      $r_{(x,y)}^{(i)} \leftarrow \hat{h}_{(x,y)}^{(i)}$   $\triangleright$  The case when  $T_{Otsu}^{(i)} \leq P_{max}^{(i)}$ 
    defined in equation (10) and (11).
17:   end if
18:   if  $I_{(x,y)}^{(i)} > 0$  then
19:      $r_{(x,y)}^{(i)} \leftarrow 1$   $\triangleright$  For every pixels  $p(x,y) > 0$  of the
    sub-image  $r^{(i)}$  set value 1 where  $I_{(x,y)}^{(i)} > 0$ .
20:   end if
21: end for
22: return  $R = \begin{bmatrix} r^{(1)} & r^{(2)} \\ r^{(3)} & r^{(4)} \end{bmatrix}$   $\triangleright$  Returns detected objects
    (assembled image  $R$ ).
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IV. IMPLEMENTATION AND RESULTS

A. Implementation

The method has been implemented in MATLAB version 9.1 (R2016b) with Image Processing Toolbox. The image processing was performed on Intel Core2 2.80 GHz CPU with 8GB of RAM.

B. Dataset

The effectiveness of our method was verified on a new image dataset¹ with 50 selected images from Google search engine (keyword ‘‘pavement cracks’’). From the review of current literature, there is a lack of publicly available dataset with RGB images that enables standardized way of results verification, as well as comparison of different methods. Hence, we’ve selected characteristic images ranging from

simple uniform cracks to more complex networks of cracks, under the influence of different illumination conditions and different weather conditions (i.e. after rain).

The selected images are collected with different types of cameras from various angles. Currently, it is common practice to analyze data captured with fixed position camera (front or back camera mounted on a specialized or passenger vehicle). This practice enables extraction of common features that can be used for segmentation. However, our method does not depend on extraction of these features, hence it can be used to analyze images from a variety of angles. We believe that this unstructured dataset can be used to examine the true potential of a method.

C. Results

The method was tested on 50 different images from our dataset. We selected images with different conditions, such as presence of lanes, vertical and horizontal cracks, pavement with water remains (i.e. rain), pavement with different shades of color (ranging from dark to light color), recessed asphalt pavement and concrete pavement. The results from selected images are shown in Fig. 3.

The method successfully detects and removes any presence of lanes in an image, even in difficult situations when lane is less visible (Fig. 3. image 1.1) or when it appears very thin due to camera angle (image 1.3b). This is enabled by application of conditions described in equation 9. The results are very promising when there are water remains on the asphalt surface. In this case, the surrounding asphalt pixels appear darker than the non-cracks pixels, hence it is very challenging to detect cracks pixels accurately. Our method detects this conditions and successfully selects only cracks pixels (image 1.1, 2.3, 3.3, 4.1, 5.3). This is enabled by applying conditions described in the equations (5) and (6). One drawback after applying these conditions is unwanted noise in the resulting images for some cases. The noise can be removed by applying simple noise-removal filter in the additional post-processing step.

Different color of the asphalt surface does not affect detection capabilities. For example, images 1.3a and 4.3 show comparable result to any other lighter color asphalt conditions. Color disbalance (image 4.1) and shadows (image 7.3) also do not have any significant impact on the resulting image. In the image 3.1 with concrete pavement, cracks have been detected with the same effectiveness as the images with asphalt pavement. In the remaining images, we showed that the method can achieve promising results whether the image has one continuous crack, a couple of interconnected horizontal and vertical cracks, or a network of many cracks.

The average execution time on the given dataset for average image size (580×800) was 338.97 ms. The method has low computational cost when compared to supervised methods and it is fast.

In Table I, we provide quantitative results for our method. The method’s accuracy has been evaluated based on Precision, Sensitivity and F-Score. We compare our results to pixel-based results. Our method shows significant improvements when compared to all unsupervised methods in [17], especially for cases when there are water remains present in asphalt surface pavements (shown as bold in Table I).

¹The dataset is available at http://people.etf.unsa.ba/~aakagic/cracks_detection/

TABLE I: Quantitative analysis of results.

	Water Remains	Dry Condition	Average
Sensitivity (Recall)	77.82%	77.73%	77.77%
Specificity	98.62%	98.97%	98.79%
Precision	77.72%	76.83%	77.27%
Accuracy	97.51%	96.75%	97.13%
F-Score	77.70%	74.49%	76.09%

V. CONCLUSION

In the recent years, computer vision has accelerated innovation process across multiple industries, from self-driving cars, advanced robotics, surveillance to facial recognition software. Development of new methods for various scientific fields is needed to achieve automatic visual understanding of useful information from a single image or a sequence of images. For public infrastructures visual inspection, it is common practice to collect data automatically with specialized vehicles, and to evaluate asphalt surface characteristics and conditions manually. Hence, time span between acquisition of data and extraction of useful information from this data can be up to several months. This is very undesirable.

In this paper, we proposed new unsupervised vision-based method for automated pavement cracks detection based on analysis of photometric information from 2D images. The method was tested on 50 images with different selection criteria. The results indicate that the method can be used for rough estimation of cracks on asphalt surface pavement. It is unsupervised method, hence it does not require existence of great amount of data needed for training. The method is simple and efficient. Its execution time is very fast. Satisfactory performance is achieved, especially in the case of low signal-to-noise ratio images.

REFERENCES

- [1] T. Wang, K. Gopalakrishnan, A. K. Somani, O. G. Smadi, and H. Ceylan, "Machine-vision-based roadway health monitoring and assessment: Development of a shape-based pavement-crack-detection approach," 2016.
- [2] S. Chambon and J.-M. Moliard, "Automatic road pavement assessment with image processing: review and comparison," *International Journal of Geophysics*, vol. 2011, 2011.
- [3] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sesselmann, D. Ebersbach, U. Stoekert, and H.-M. Gross, "How to get pavement distress detection ready for deep learning? a systematic approach," in *Neural Networks (IJCNN)*, 2017 *International Joint Conference on*, pp. 2039–2047, IEEE, 2017.
- [4] E. Buza, S. Omanovic, and A. Huseinovic, "Pothole detection with image processing and spectral clustering," in *Proceedings of the 2nd International Conference on Information Technology and Computer Networks*, pp. 48–53, 2013.
- [5] A. Akagic, E. Buza, and S. Omanovic, "Pothole detection: An efficient vision based method using rgb color space image segmentation," in *Information and Communication Technology, Electronics and Micro-electronics (MIPRO)*, 2017 *40th International Convention on*, pp. 1104–1109, IEEE, 2017.
- [6] E. Buza, A. Akagic, S. Omanovic, and H. Hasic, "Unsupervised method for detection of high severity distresses on asphalt pavements," in *Informatics*, 2017 *IEEE 14th International Scientific Conference on*, pp. 45–50, IEEE, 2017.
- [7] J.-B. Ihn and F.-K. Chang, "Detection and monitoring of hidden fatigue crack growth using a built-in piezoelectric sensor/actuator network: II. validation using riveted joints and repair patches," *Smart materials and structures*, vol. 13, no. 3, p. 621, 2004.
- [8] R. Amhaz, S. Chambon, J. Idier, and V. Baltazart, "Automatic crack detection on two-dimensional pavement images: an algorithm based on minimal path selection," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 10, pp. 2718–2729, 2016.
- [9] L. Zhang, F. Yang, Y. D. Zhang, and Y. J. Zhu, "Road crack detection using deep convolutional neural network," in *Image Processing (ICIP)*, 2016 *IEEE International Conference on*, pp. 3708–3712, IEEE, 2016.
- [10] Q. Li, D. Zhang, Q. Zou, and H. Lin, "3d laser imaging and sparse points grouping for pavement crack detection," in *Signal Processing Conference (EUSIPCO)*, 2017 *25th European*, pp. 2036–2040, IEEE, 2017.
- [11] V. Ananth, P. Ananthi, V. Elakkiya, J. Priyadarshini, and R. Shiyamili, "Automatic pavement crack detection algorithm," *International Innovative Research Journal of Engineering and Technology*, vol. 2, 2017.
- [12] H. Oliveira and P. L. Correia, "Supervised strategies for cracks detection in images of road pavement flexible surfaces," in *Signal Processing Conference, 2008 16th European*, pp. 1–5, IEEE, 2008.
- [13] J. Bray, B. Verma, X. Li, and W. He, "A neural network based technique for automatic classification of road cracks," in *Neural Networks, 2006. IJCNN'06. International Joint Conference on*, pp. 907–912, IEEE, 2006.
- [14] A. Cord and S. Chambon, "Automatic road defect detection by textural pattern recognition based on adaboost," *Computer-Aided Civil and Infrastructure Engineering*, vol. 27, no. 4, pp. 244–259, 2012.
- [15] P. Subirats, J. Dumoulin, V. Legeay, and D. Barba, "Automation of pavement surface crack detection with a matched filtering to define the mother wavelet function used," in *Signal Processing Conference, 2006 14th European*, pp. 1–5, IEEE, 2006.
- [16] R. Amhaz, S. Chambon, J. Idier, and V. Baltazart, "A new minimal path selection algorithm for automatic crack detection on pavement images," in *Image Processing (ICIP)*, 2014 *IEEE International Conference on*, pp. 788–792, IEEE, 2014.
- [17] Y. Shi, L. Cui, Z. Qi, F. Meng, and Z. Chen, "Automatic road crack detection using random structured forests," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 12, pp. 3434–3445, 2016.
- [18] H. Oliveira and P. L. Correia, "Automatic road crack segmentation using entropy and image dynamic thresholding," in *Signal Processing Conference, 2009 17th European*, pp. 622–626, IEEE, 2009.
- [19] H. Oliveira and P. L. Correia, "Automatic road crack detection and characterization," *IEEE Transactions on Intelligent Transportation Systems*, vol. 14, no. 1, pp. 155–168, 2013.
- [20] J. A. Acosta, J. L. Figueroa, and R. L. Mullen, "Low-cost video image processing system for evaluating pavement surface distress," *Transportation research record*, no. 1348, 1992.
- [21] J. Tang and Y. Gu, "Automatic crack detection and segmentation using a hybrid algorithm for road distress analysis," in *Systems, Man, and Cybernetics (SMC)*, 2013 *IEEE International Conference on*, pp. 3026–3030, IEEE, 2013.
- [22] M. Gavilán, D. Balcones, O. Marcos, D. F. Llorca, M. A. Sotelo, I. Parra, M. Ocaña, P. Aliseda, P. Yarza, and A. Amírola, "Adaptive road crack detection system by pavement classification," *Sensors*, vol. 11, no. 10, pp. 9628–9657, 2011.
- [23] H. Koutsopoulos and A. Downey, "Primitive-based classification of pavement cracking images," *Journal of Transportation Engineering*, vol. 119, no. 3, pp. 402–418, 1993.
- [24] N. Coudray, A. Karathanou, and S. Chambon, "Multi-resolution approach for fine structure extraction: Application and validation on road images," in *International Joint Conference on Computer Vision Theory and Applications, VISAPP*, p. sp, 2010.
- [25] P. Delagnes and D. Barba, "A markov random field for rectilinear structure extraction in pavement distress image analysis," in *Image Processing, 1995. Proceedings., International Conference on*, vol. 1, pp. 446–449, IEEE, 1995.
- [26] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, "Cracktree: Automatic crack detection from pavement images," *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.
- [27] S. Chambon, "Detection of points of interest for geodesic contours: Application on road images for crack detection," in *International Joint Conference on Computer Vision Theory and Applications, VISAPP*, p. sp, 2011.
- [28] T. S. Nguyen, S. Begot, F. Duculty, and M. Avila, "Free-form anisotropy: A new method for crack detection on pavement surface images," in *Image Processing (ICIP)*, 2011 *18th IEEE International Conference on*, pp. 1069–1072, IEEE, 2011.
- [29] A. Mohan and S. Poobal, "Crack detection using image processing: A critical review and analysis," *Alexandria Engineering Journal*, 2017.

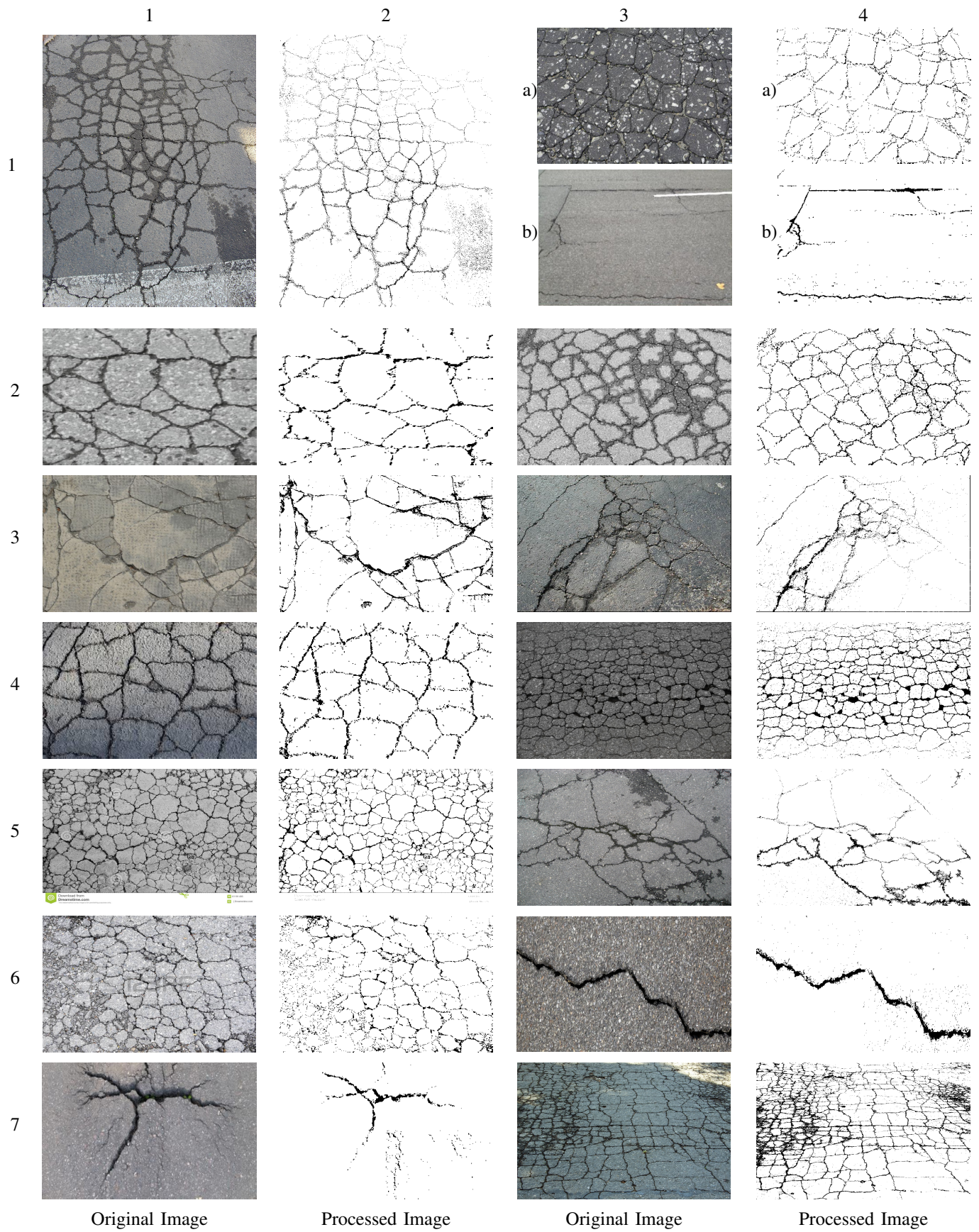


Figure 3: Cracks detection capabilities of our method showing *Original* and *Processed* images - detected cracks.